Examiner only

9. A student aims to improve his fitness.
Starting from rest, he can accelerate at $a = 2\,\text{m/s}^2$ for a time, $t = 4\,\text{s}$.

Use the information above to answer the following questions.

(a) (i) State the value of the initial velocity, u. [1]

initial velocity, u = ... m/s

(ii) Use the equation:

$$v = u + at$$

to calculate his final velocity, v. [2]

final velocity, v = ... m/s

(b) Use the equation:

$$x = \frac{(u + v)}{2}\ t$$

to calculate the distance travelled, x, in this time of 4 s. [2]

distance, x = ... m

(c) The student thinks that there are two ways that he could increase his acceleration.
1. Reducing body mass.
2. Increasing the strength of his legs.

Explain whether you agree with his ideas. Use $a = \frac{F}{m}$. [2]

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